

Charlie Pearson

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Education

University of Edinburgh – Meng Mechanical Engineering (2023 – 2028)

- First year class medal, awarded to best performing student in the cohort
- Year-long exchange at Cornell University (3rd year) with 3.78 GPA
- Predicted 1st class honours
- Relevant Modules: Solid Mechanics, System Dynamics, Engineering Design, Material Processing, Thermodynamics, Fluid Mechanics, Numerical Methods

The Sixth Form College Farnborough (2020-2022)

- A* in Mathematics, Physics and Economics
- President of the engineering society

Experience

BUiD Sustainability Spring School (2025)

- Collaborated with engineers, architects, and designers to develop a sustainable lighting solution, integrating mechanical design and energy efficiency principles
- Pitched the final design to a panel of engineers and academics, achieving 1st place
- Communicated technical concepts within an international team, supporting members for whom English was not a first language

CFH Tutoring, Co-Founder and Tutor (2020-2025)

- Co-founded a tutoring company educating and mentoring children from ages 8-18
- Grew the business to a revenue of over £15,000 and employed up to 8 tutors
- Developed a structured curriculum to ensure quality and consistency, yielding a 100% pass rate over a 5-year period

Costa Coffee, Barista (2022-Present)

- Led shifts of 4–6 staff in a fast-paced environment fostering teamwork and efficiency
- Implemented stock management strategies, reducing monthly waste by 15%

Chillisauc Ltd, Work Experience Week (2019)

- Gained broad insight into business operations through departmental rotations and project shadowing

Projects

Instrumented Torque Wrench Design

- Designed a 600 in-lbf torque wrench with integrated strain gauge instrumentation, optimising for sensitivity while meeting fatigue life and fracture safety requirements
- Performed materials selection and iterative design using Python scripts and Granta to balance competing constraints (yield strength, fracture toughness, fatigue resistance)
- Validated hand calculations against ANSYS FEA models, achieving >1.0 mV/V sensor output while maintaining required safety factors
- Created parametric CAD model in Fusion 360 and conducted mesh refinement studies to verify stress concentrations

Technical Skills

Engineering & Analysis

- Structural analysis using finite element methods in ANSYS
- Material selection and optimisation using Granta EduPack
- Engineering calculations and design validation

CAD & Design

- Fusion 360 - parametric modelling, assemblies and design iteration
- AutoCAD - 2D engineering drawings

Programming & Computational Tools

- Python – primary language for engineering scripting and numerical analysis
- MATLAB – foundational experience from engineering coursework

Laboratory & Manufacturing

- Operation of mill, lathe, and bandsaw for prototype fabrication
- Mechanical testing using tensile testing machines in accordance with ASTM standards
- Oscilloscope analysis of free and forced vibration response in dynamic systems

Technical Communication

- Technical reporting using LaTeX and Microsoft Office for clear communication of analysis and results

Leadership and Additional Experience

- Captained school football team to its first county cup final, developing teamwork and performance-driven leadership
- International academic and project experience through study at Cornell University and participation in sustainability programme in Dubai